



COMPUTER ARCHITECTURE

Lab Task # 7



Name: Horriya Noor

Sap Id: 64048

Semester: 4th Semester

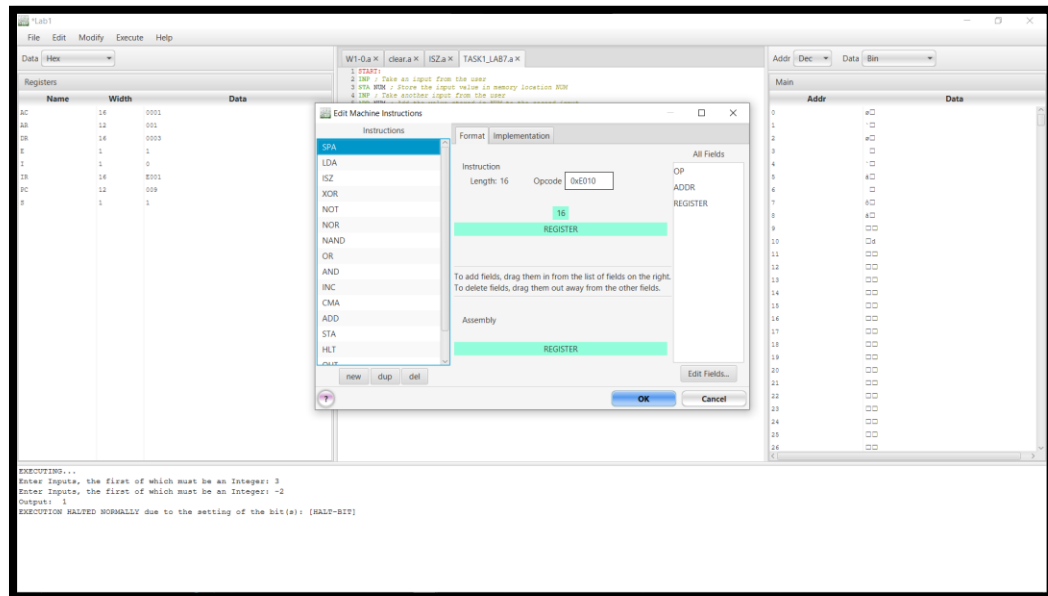
Department: BSCS

Lab Task-1 Task

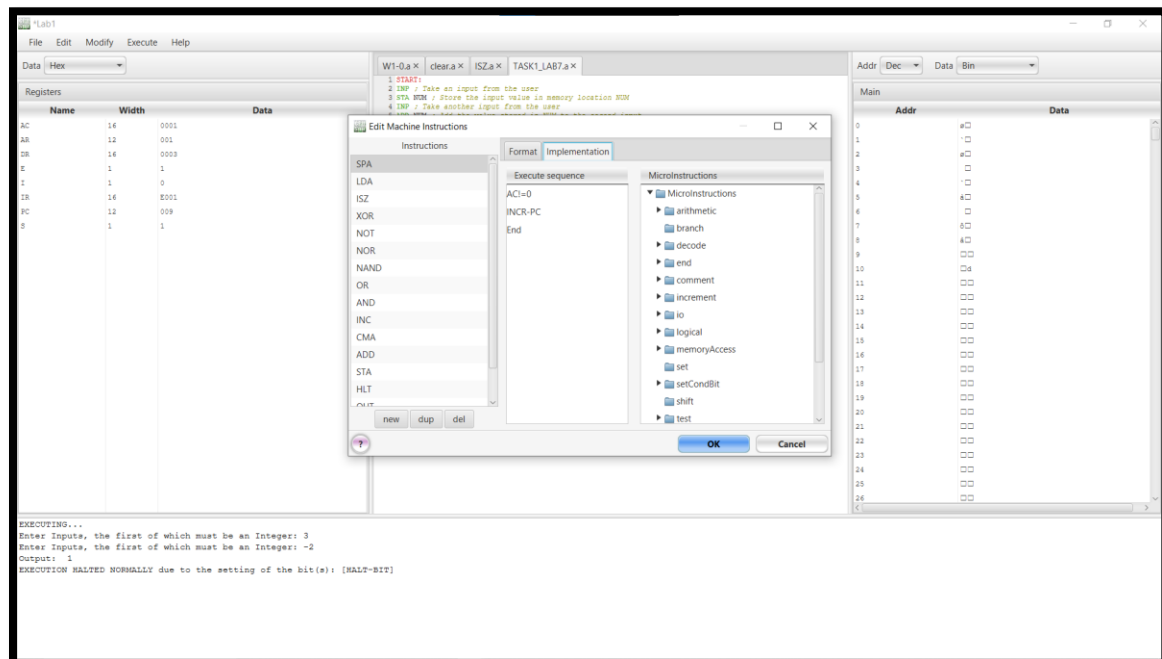
Requirements:

1. Design a machine to solve the Program-1 given at start of the document.

Instructions Format:



Instructions Implementation:



Result (Using Both Positive Numbers)

The screenshot shows the 'Lab1' window with the following components:

- Registers:** A table showing the state of various registers. The 'Data' column for the 'SP' register is 004B.
- Assembly Code:** A list of instructions for 'TASK1_LAB7.a'. The code includes instructions for taking input, storing it in memory, adding values, and outputting the result.
- Memory:** A table showing memory locations from 0 to 26. The 'Data' column for address 0 is 004B.
- Execution Log:** A text area at the bottom showing the execution process. It indicates that the program executed normally due to the setting of the bit(s): [HALT-BIT].

EXECUTING...

Enter Inputs, the first of which must be an Integer: 3

Enter Inputs, the first of which must be an Integer: 4

Output: 107

EXECUTION HALTED NORMALLY due to the setting of the bit(s): [HALT-BIT]

Result(Using one Positive and other negative Number):

The screenshot shows the 'Lab1' window with the following components:

- Registers:** A table showing the state of various registers. The 'Data' column for the 'SP' register is 004B.
- Assembly Code:** A list of instructions for 'TASK1_LAB7.a'. The code includes instructions for taking input, storing it in memory, adding values, and outputting the result.
- Memory:** A table showing memory locations from 0 to 26. The 'Data' column for address 0 is 004B.
- Execution Log:** A text area at the bottom showing the execution process. It indicates that the program executed normally due to the setting of the bit(s): [HALT-BIT].

EXECUTING...

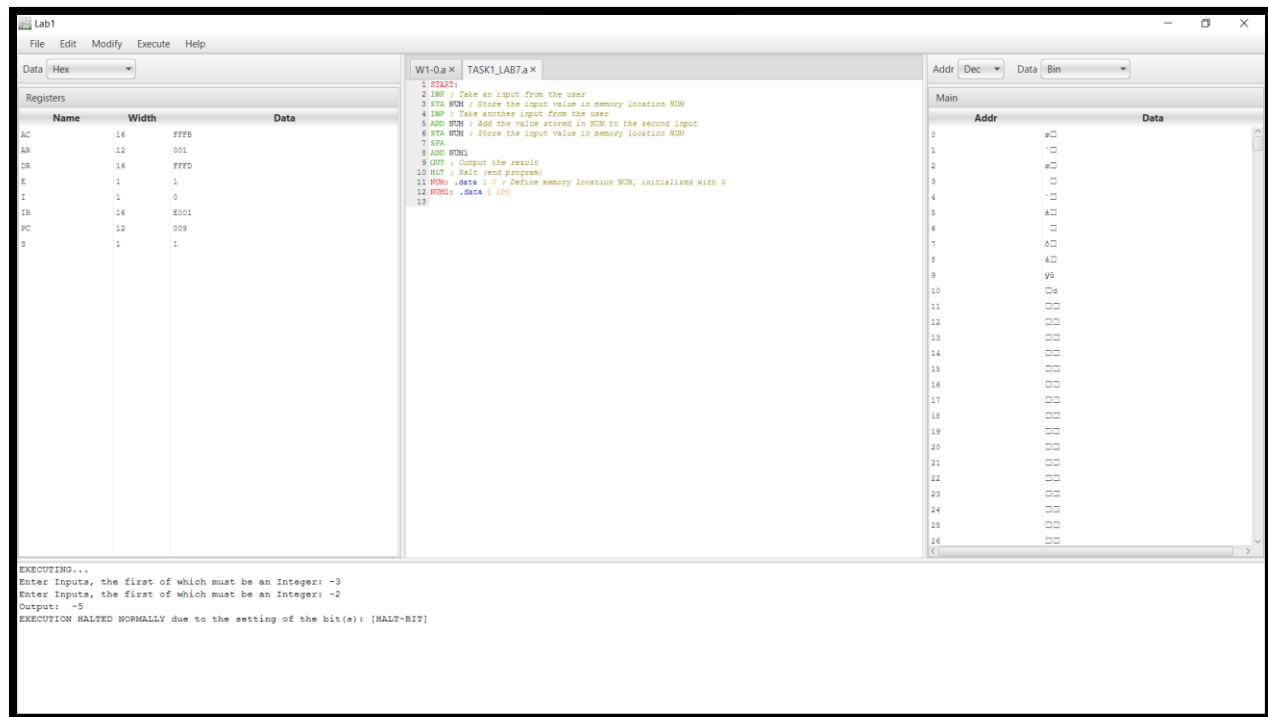
Enter Inputs, the first of which must be an Integer: 3

Enter Inputs, the first of which must be an Integer: -2

Output: 1

EXECUTION HALTED NORMALLY due to the setting of the bit(s): [HALT-BIT]

Result (Using both negative Number):



Explanation:

The number 1 has value 2 (first input) and number 2 has value 3 (second input) taken from the user by using INP and store in Accumulator Register (AC). The first input from AC is stored in memory location. INP takes the second input from the user and loads it into the accumulator. Then it adds the value stored in NUM (first input) to the second input. After that, the STA NUM stores the result of the addition back into memory location NUM. SPA (Skip if Positive) commands checks the sign of the accumulator and verify the conditions that:

- If the value is **positive**, the next instruction is skipped.
- If the value is **negative**, the next instruction executes.

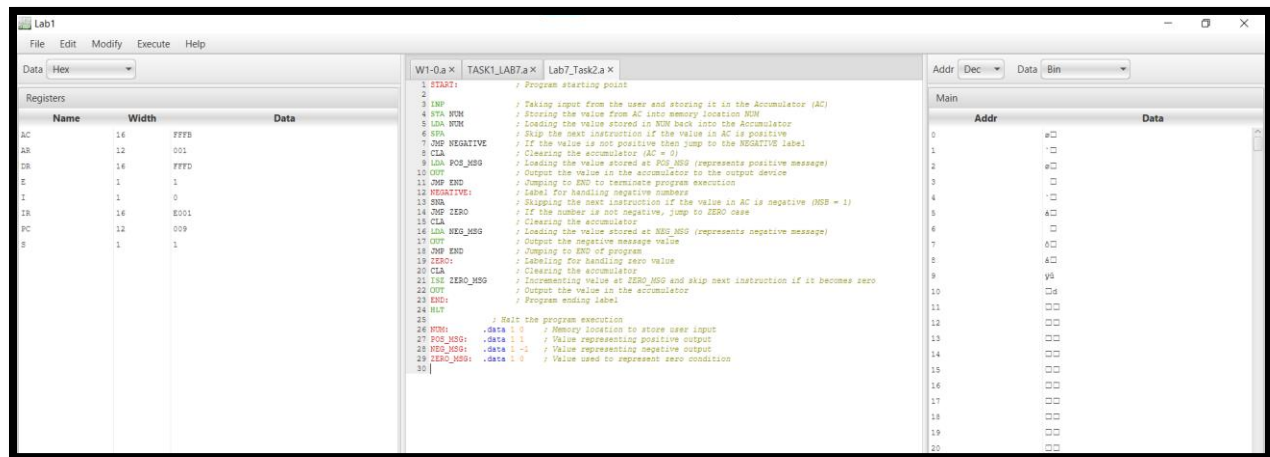
After verifying the condition, if the previous result was negative. 100 will be added to accumulator. OUT instruction will display the value stored in the accumulator. And lastly, HLT will stop the execution of program.

Conclusion:

- The program **adds two numbers**.
- If the result is **negative**, it **adds 100** before displaying the output.
- If the result is **positive**, it directly displays the result.

Lab Task-2

Read the following code, write explanation of each line in front of each line. Implement this code using CPUSim submit screenshot.



Conclusion:

This program checks that the value entered by the user and determines whether it is **positive, negative, or zero** using the instructions SPA (Skip if Positive) and SNA (Skip if Negative). Based on the condition, the program jumps to the corresponding section and outputs the appropriate value: **1 for positive, -1 for negative, and 0 for zero.**